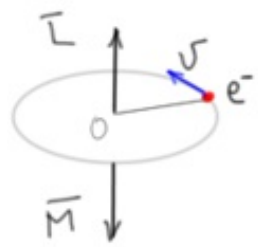


El momentum angular y el momento dipolar magnético



$$\vec{L} = \vec{r} \times \vec{p}$$

Momentum Angular

$$\vec{M} = IA \hat{n}$$

Momento dipolar magnético

$$\vec{L} = \mu r v \hat{n}, \quad \vec{M} = -\pi r^2 \frac{|e| v}{2\pi r} \hat{n}$$

$$= -\frac{|e|}{2\mu} \vec{L}$$

$$\vec{M} = -\frac{|e|}{2\mu} \vec{L}$$

Por la ley de la conservación de la carga:

$$\nabla \cdot \vec{J}_e + \partial \rho_e = 0$$

Densidad de corriente eléctrica Densidad de carga eléctrica

Hacemos:

$$\vec{J}_e = e \vec{J}$$

$$\rho_e = e \rho$$

$$e \nabla \cdot \vec{J} + \partial \rho = 0$$

$$\leftrightarrow \nabla \cdot \vec{J}_e + \partial \rho_e = 0$$

$$\rho = |\psi|^2; \quad \vec{J} = \frac{1}{\mu} \text{Re} [\psi^* \vec{p} \psi]$$

$$\vec{p} = \frac{\hbar}{i} \left[\hat{e}_r \mathcal{D}_r + \frac{1}{r} \hat{e}_\theta \mathcal{D}_\theta + \frac{1}{r \sin \theta} \hat{e}_\phi \mathcal{D}_\phi \right]$$

$$\psi = \eta_{Nl} Y_{lm} = G_{Nl} r^l e^{-6r} Y_{lm}$$

$$\bullet \psi^* D_r \psi \in \mathbb{R}$$

$$\eta^* Y^* D_r \eta Y = \eta^* D_r \eta \in \mathbb{R}$$

$$\bullet \psi^* D_\theta \psi \in \mathbb{R}$$

$$\eta^* Y^* D_\theta \eta Y = Y^* D_\theta Y \in \mathbb{R}$$

$$(Y = S_{lm}(\theta) e^{im\phi})$$

$$\bullet \psi^* D_\phi \psi = im \psi^* \psi$$

$$(D_\phi \psi = im \psi)$$

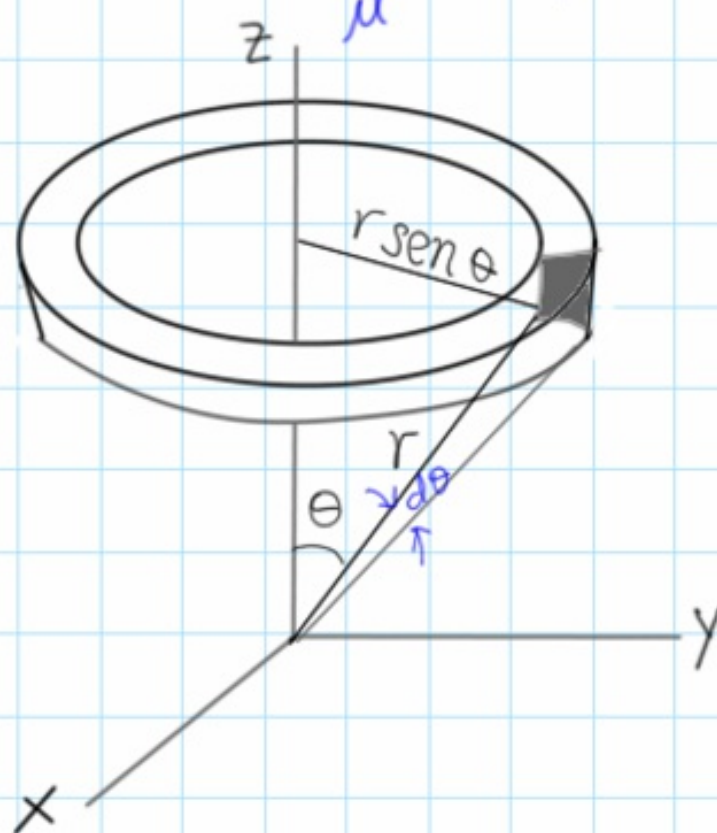
Verificar que

$$\bar{J} = \frac{\hbar m}{\mu r \sin \theta} |\psi|^2 \hat{e}_\phi$$

$$\bar{p} \psi = \frac{\hbar}{i} \left[\hat{e}_r D_r \psi + \frac{1}{r} \hat{e}_\theta D_\theta \psi + \frac{1}{r \sin \theta} \hat{e}_\phi D_\phi \psi \right]$$

$$\text{Re } \psi^* \bar{p} \psi = \frac{\hbar}{i} \frac{1}{r \sin \theta} \hat{e}_\phi \psi^* D_\phi \psi = \frac{\hbar m}{r \sin \theta} |\psi|^2 \hat{e}_\phi$$

Como $\bar{J} = \frac{1}{\mu} \text{Re } \psi^* \bar{p} \psi$, se verifica lo que se pide.



Podemos interpretar este resultado como la densidad de corriente que circula alrededor de z .

$$d\bar{a} = r d\theta dr \hat{e}_\phi$$

$$d\bar{I} = \bar{J}_e \cdot d\bar{a} = \frac{e \hbar m r}{\mu r \sin \theta} |\psi|^2 d\theta dr$$

$$= \frac{e \hbar m |\psi|^2}{\mu \sin \theta} d\theta dr$$

$$d\bar{M} = \frac{1}{c} \pi (r \sin \theta)^2 \hat{e}_3 dI$$

$$= -m M_B \hat{e}_3 2\pi r^2 |\psi|^2 \sin \theta dr d\theta$$

$$M_B = \frac{|e| \hbar}{2\mu c} \quad \text{Magnetón de Bohr}$$

Encontremos $\bar{M}$

$$\bar{M} = -m M_B \hat{e}_3 \underbrace{\int_0^\pi \int_0^\infty |\psi|^2 2\pi r^2 \sin \theta dr d\theta}_1$$

$$\bar{M} = -m M_B \hat{e}_3 \quad \text{Momento magnético orbital}$$

$$L_3 \psi = m \hbar \psi$$

